

## Algebra

|                                               |                                                |
|-----------------------------------------------|------------------------------------------------|
| $(a \pm b)^2 = a^2 \pm 2ab + b^2$             | $(a + b)(a - b) = a^2 - b^2$                   |
| $(a \pm b)^3 = a^3 \pm 3a^2b + 3ab^2 \pm b^3$ | $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$          |
| $ax^2 + bx + c = a(x - x_1)(x - x_2)$         | $x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ |

|                                 |                                                          |
|---------------------------------|----------------------------------------------------------|
| $a^x \cdot b^x = (a \cdot b)^x$ | $\frac{a^x}{b^x} = \left(\frac{a}{b}\right)^x$           |
| $a^x \cdot a^y = a^{x+y}$       | $\log_a(x \cdot y) = \log_a(x) + \log_a(y)$              |
| $\frac{a^x}{a^y} = a^{x-y}$     | $\log_a\left(\frac{x}{y}\right) = \log_a(x) - \log_a(y)$ |
| $(a^x)^y = a^{(x \cdot y)}$     | $\log_a(x^y) = y \cdot \log_a(x)$                        |
| $a^{\log_a(y)} = y$             | $\log_a(a^x) = x$                                        |

## Trigonometry

|                                                       |                                                       |                                    |
|-------------------------------------------------------|-------------------------------------------------------|------------------------------------|
| $\cos(-x) = \cos(x)$                                  | $\sin(-x) = -\sin(x)$                                 | $\cos^2(x) + \sin^2(x) = 1$        |
| $\sin\left(\frac{\pi}{2} + x\right) = \cos(x)$        | $1 + \tan^2(x) = \frac{1}{\cos^2(x)}$                 | $\sin(2x) = 2 \sin(x) \cos(x)$     |
| $\sin(x \pm y) = \sin(x) \cos(y) \pm \sin(y) \cos(x)$ | $\cos(x \pm y) = \cos(x) \cos(y) \mp \sin(x) \sin(y)$ | $\cos(2x) = \cos^2(x) - \sin^2(x)$ |
| $\sin(x \pm 2\pi k) = \sin(x)$                        | $\cos(x \pm 2\pi k) = \cos(x)$                        | $\tan(x \pm \pi k) = \tan(x)$      |

$$\cos(x) + \cos(y) = 2 \cos\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) \quad \left| \quad \sin(x) + \sin(y) = 2 \sin\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) \right|$$

|                                                                                        |                                                                                 |                                                                                        |                                                |
|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------|
| $\sin\left(\frac{\pi}{3}\right) = \cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$ | $\sin\left(\frac{\pi}{6}\right) = \cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$ | $\sin\left(\frac{\pi}{4}\right) = \cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$ | $\sin\left(\frac{\pi}{2}\right) = \cos(0) = 1$ |
| $\cos\left(\frac{\pi}{2}\right) = \sin(0) = 0$                                         | $\tan\left(\frac{\pi}{3}\right) = \sqrt{3}$                                     | $\tan\left(\frac{\pi}{6}\right) = \frac{1}{\sqrt{3}}$                                  | $\tan\left(\frac{\pi}{4}\right) = 1$           |

|                                                                                                                               |                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| $\arcsin(y) = x \leftrightarrow \sin(x) = y \quad \forall -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}, \forall -1 \leq y \leq 1$ | $\arccos(y) = x \leftrightarrow \cos(x) = y \quad \forall 0 \leq x \leq \pi, \forall -1 \leq y \leq 1$ |
| $\arctan(y) = x \leftrightarrow \tan(x) = y \quad \forall -\frac{\pi}{2} < x < \frac{\pi}{2}, \forall y \in \mathbb{R}$       |                                                                                                        |

## AOL

|                                  |                                   |
|----------------------------------|-----------------------------------|
| $\infty + \infty = \infty$       | $\infty \cdot \infty = \infty$    |
| $\frac{1}{\infty} = 0$           | $\infty + c = \infty$             |
| $c \cdot \infty = \infty, c > 0$ | $c \cdot \infty = -\infty, c < 0$ |
| $\frac{1}{0^+} = \infty$         | $\frac{1}{0^-} = -\infty$         |
| $\sqrt{\infty} = \infty$         | $\infty^\infty = \infty$          |
| $\infty^c = \infty, c > 0$       | $\infty^c = 0, c < 0$             |

## Integration and differentiation

|                                                                                                       |                                                                            |
|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| $\int x^\alpha dx = \frac{x^{\alpha+1}}{\alpha+1} + C, (\alpha \neq -1)$                              | $(x^\alpha)' = \alpha x^{\alpha-1}$                                        |
| $\int \frac{1}{x} dx = \ln  x  + C$                                                                   | $(\ln x)' = \frac{1}{x}$                                                   |
| $\int \cos x dx = \sin x + C$                                                                         | $(\sin x)' = \cos x$                                                       |
| $\int \sin x dx = -\cos x + C$                                                                        | $(\cos x)' = -\sin x$                                                      |
| $\int \frac{1}{\cos^2 x} dx = \tan x + C$                                                             | $(\tan x)' = \frac{1}{\cos^2 x}$                                           |
| $\int a^x dx = \frac{a^x}{\ln a} + C$                                                                 | $(a^x)' = \ln a \cdot a^x$                                                 |
| $\int e^x dx = e^x + C$                                                                               | $(e^x)' = e^x$                                                             |
| $\int \frac{1}{1+x^2} dx = \arctan x + C$                                                             | $(\arctan x)' = \frac{1}{1+x^2}$                                           |
| $\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$                                                      | $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$                                    |
| $\int \frac{1}{-\sqrt{1-x^2}} dx = \arccos x + C$                                                     | $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$                                   |
| $\int \frac{1}{a^2+x^2} dx = \frac{1}{a} \arctan \frac{x}{a} + C$                                     | $(\log_a x)' = \frac{1}{\ln(a) \cdot x}$                                   |
| $\int \frac{1}{\sqrt{a^2-x^2}} dx = \arcsin \frac{x}{a} + C$                                          |                                                                            |
| $\int \alpha \cdot f(x) dx = \alpha \cdot \int f(x) dx$                                               | $[\alpha \cdot f(x)]' = \alpha \cdot f'(x)$                                |
| $\int f(x) \pm g(x) dx = \int f(x) dx \pm \int g(x) dx$                                               | $[f(x) \pm g(x)]' = f'(x) \pm g'(x)$                                       |
| $\int u'(x)v(x) dx = u(x)v(x) - \int u(x)v'(x) dx$                                                    | $[u(x) \cdot v(x)]' = u'(x)v(x) + u(x)v'(x)$                               |
| $\int g(f(x))f'(x) dx = \int g(t) dt$                                                                 | $[g(f(x))]' = g'(f(x)) \cdot f'(x)$                                        |
| $\int_1^\infty \frac{dx}{x^p} = \begin{cases} \frac{1}{p-1} & p > 1 \\ \infty & p \leq 1 \end{cases}$ | $\left[ \frac{f(x)}{g(x)} \right]' = \frac{f'(x)g(x) - g'(x)f(x)}{g^2(x)}$ |
| $\int_0^1 \frac{dx}{x^p} = \begin{cases} \frac{1}{1-p} & p < 1 \\ \infty & p \geq 1 \end{cases}$      | $[f^{-1}(y)]' = \frac{1}{f'(f^{-1}(y))}$                                   |

## Power series

|                                                                                                                              |                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| $T_f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n$                                                           | $R_N(x) = f(x) - T_N(x) = \frac{f^{(N+1)}(\xi)}{(N+1)!} (x - x_0)^{N+1}$                                                  |
| $R = \frac{1}{\lim_{n \rightarrow \infty} \sqrt[n]{ a_n }} = \lim_{n \rightarrow \infty} \left  \frac{a_n}{a_{n+1}} \right $ | $R = \frac{1}{\lim_{n \rightarrow \infty} \frac{ a_{n+1} }{ a_n }} = \lim_{n \rightarrow \infty} \frac{ a_n }{ a_{n+1} }$ |
| $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n,  x  < 1$                                                                           | $\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}, -1 < x \leq 1$                                                  |
| $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, x \in \mathbb{R}$                                                                 | $\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, x \in \mathbb{R}$                                          |
| $\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, x \in \mathbb{R}$                                                 | $\arctan x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}, -1 \leq x \leq 1$                                          |